

AI-Driven Pay Equity Analytics: Detecting and Addressing Compensation Disparities

FIRST DRAFT

Thesis Paper Proposal

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1. Introduction

1.1 Introduction to the Topic

1.2 Research Motivation

1.3 Research Gap

1.4 Research Objectives

1.5 Research Methodology Overview

1.6 Thesis Structure

2. *Literture Review*

- **2.1 Foundations of Pay Equity and Algorithmic Analysis**

2.1.1 Gender Pay Gap and Persistent Disparities

2.1.2 Limitations of Traditional Pay Analysis Methods

2.1.3 Emergence of AI in Compensation Systems

2.1.4 Key Fairness Concepts in Pay Equity Analytics

- **2.2 AI Approaches for Detecting Compensation Disparities**

2.2.1 Machine Learning Models for Wage Prediction and Analysis

2.2.2 Bayesian and Probabilistic Models in Pay Equity

2.2.3 Algorithmic Fairness Techniques in Compensation Systems

2.3 Algorithmic Bias, Data Quality, and Equity Risks

2.3.1 Bias Embedded in Compensation Data

2.3.2 Dataset Auditability and Quality Challenges

2.3.3 Discrimination Risks Posed by Algorithmic Wage-Setting

2.4 AI Techniques for Detecting Compensation Disparities

2.4.1 Machine Learning Wage Prediction Models

2.4.2 Bayesian and Probabilistic Fairness Models

2.4.3 AI Fairness and Counterfactual Techniques

2.5 AI Methods for Correcting Compensation Inequities

2.5.1 Fairness-Constrained Compensation Optimization

2.5.2 Pre-Processing Bias Reduction Techniques

2.5.3 In-Processing and Post-Processing Fairness Corrections

2.6 Practical AI Applications for Achieving Pay Equity

2.6.1 Real-Time Pay Equity Monitoring and Dashboards

2.6.2 Automated Compensation Correction Systems

2.6.3 Scenario Simulation and Policy Testing

2.6.4 Integration with Governance, Legal Requirements, and Ethical Standards

2.7- Conclusion

3. Research Methodology

3.1 Research Design: Design Science Approach

3.2 Design Science Process Steps 3.2.1 Problem Identification

3.2.2 Objective Definition

3.2.3 Design and Development

3.2.4 Demonstration

3.2.5 Evaluation

3.2.6 Communication

3.3 Justification for Using Design Science

4. System Design and Implementation

4.1 System Overview

4.1.1 Model Selection and Justification

4.2 Data Processing and Feature Engineering Dataset Structure Cleaning Features

4.3 System Functional Modules

4.3.1 Salary Prediction Module

4.3.2 Inequity Detection Module

4.3.3 Pay Gap Analysis

4.3.4 Bias Detection Mechanism

4.3.5 Budget Simulation Engine

4.3.6 Recommendation System

4.4 Application of Design Science (Step-by-Step)

4.4.1 Problem Identification in Practice

4.4.2 Objective Definition in System Context

4.4.3 Artifact Development

4.4.4 Demonstration Using Dataset

4.4.5 Evaluation Through System Outputs

4.4.6 Communication of Results

4.5 Expert Validation (HR Interview)

5. Discussion

5.1 Interpretation of Results What did your system discover

5.2 Addressing the Research Problem How your system solves pay inequity

5.3 Comparison with Traditional Methods Why your system is better

5.4 System Strengths transparency automation fairness

5.5 System Limitations dataset size assumptions

6. Conclusion

6.1 Summary of Findings

6.2 Theoretical Implications

6.3 Practical Implications

6.4 Recommendations

6.5 Limitations & Future Work

1. Introduction

1.1 Introduction to the Topic

Artificial intelligence is changing how companies make decisions, especially when it comes to things like hiring, performance reviews, and pay. In the past, decisions about compensation usually relied on managers' gut feelings or simple statistical models, which really didn't do a good job capturing just how complicated salary structures actually are. Now, with data-driven technology, organizations are turning to AI to make sense of pay patterns, aiming to be more accurate and fair. These AI systems can pull patterns from huge amounts of employee data, figuring out links between things like experience, job role, and performance, so salary decisions aren't just more objective—they're better informed.

That all sounds promising, but using AI for compensation brings its own mix of ups and downs. Sure, AI can spot hidden pay gaps and let organizations keep a much closer eye on how salaries match up across the board. But here's the catch: if the data you feed the system already carries traces of old biases—like those about gender or structural inequality—the algorithms can easily repeat those mistakes. That's why it's so important to design these systems with fairness and transparency front and center. This research is all about pay equity analytics fueled by AI. The goal? Use machine learning and fairness-focused tech to spot and address pay disparities so organizations can actually do something about them.

1.2 Research Motivation

Pay equity is still a real problem for companies everywhere, even though more people are talking about it and there's pressure to fix it. Gender pay gaps and other forms of unfair pay keep turning up, even after you account for things like role, experience, and how well people do their jobs. Most of these gaps come from deeper, structural issues that are tough to spot using old-school methods, which often depend on oversimplified assumptions. These traditional tools just can't handle the complicated nature of how compensation actually works today.

That's where AI steps in. Companies want decisions that are scalable, dependable, and rooted in data. Machine learning can sift through pay data, surfacing employees whose compensation looks out of line with expectations. This gives organizations a way to find and fix inequities before they become bigger problems. But, let's be honest, depending on algorithms isn't totally

smooth sailing—bias, opacity, and ethical issues crop up fast. If you don't build in the right checks, AI can end up making things worse, not better. That's why this research aims to build an AI system that doesn't just point out pay gaps but also explains why they exist and what organizations can do to close them, offering solutions that are transparent and practical.

1.3 Research Gap

Even though there's plenty of research proving AI can help analyze pay data, a few important things are missing. For one, most studies just focus on finding gaps—they rarely suggest real, hands-on ways for organizations to fix them. So we know AI can spot unfair pay, but how do you actually use that info to change things inside companies?

There's another problem: the tools aren't always transparent. Many AI models work like black boxes, and HR teams are left in the dark about why certain salaries get flagged. If people don't trust the tools—or can't see how the decisions are made—they probably won't use them. Plus, while some systems might offer one or two key features, very few put everything together: real-time monitoring, bias detection, simulation, and recommendations, all in one package.

There's also not much research using artifact-based approaches, like Design Science Research, in this field. People talk about theories and analytics, sure, but very few actually build or test systems that could work in the messy reality of real organizations. This thesis addresses all these gaps by building and testing a complete AI-powered solution—one that brings together detection, explanation, simulation, and actionable recommendations in a single framework.

1.4 Research Objectives

This research has one big goal: create an AI system that helps companies spot and address pay inequities in a straightforward, transparent way. The main approach is to design a machine learning model that predicts what a fair salary should look like for each employee, based on relevant info like job role, performance, and experience. When an employee's actual pay doesn't line up with what's expected, the system points that out.

But it goes beyond just detecting gaps. The research also adds a component to spot bias—especially gender-based pay gaps—and allows HR teams to run simulations: they can see what happens if they distribute a fixed budget to adjust pay, and actually measure the fairness impact of those choices. At the end of the day, the system doesn't just offer numbers; it also provides clear recommendations, helping HR teams make smarter, fairer decisions. A major focus is on explainability. Everything the system outputs should be accessible and easy to understand so that users trust the process. Each of these objectives connects directly with the modules built into the system: salary prediction, inequity detection, fairness analysis, simulation, and recommendation.

1.5 Research Methodology Overview

This thesis uses Design Science Research (DSR)—a method tailored for tackling real-world, practical problems with new technology. DSR is all about building “artifacts”—actual solutions—then testing how well they solve the problem at hand. In this project, the artifact is an AI-driven system for fair compensation decisions, combining machine learning with analytics and HR decision support.

The process itself is pretty hands-on: identify the problem, set objectives, design your solution, build it, test it, and share what you find. The end result is an interactive app where users can upload or generate their own data, analyze salary structures, and try out different adjustments. The machine learning piece predicts what employees should be earning, and the system generates fairness metrics, like pay gaps or indications of bias. Simulations let users experiment with various budget allocation strategies to see what actually moves the needle on pay equity. The system's value comes from the outputs: concrete fairness measures, bias detection, and real recommendations.

1.6 Thesis Structure

The thesis has six chapters, each digging into a different part of the research. The first chapter sets the stage, laying out the motivation, gaps, goals, and the research approach. The second chapter pulls together the literature on pay equity, AI for compensation analysis, algorithmic fairness, and bias mitigation. The third chapter dives deeper into Design Science Research and explains how it's used in this project. In the fourth chapter, you'll find all the details about how the system was built—covering model development, data processing, and practical features. The fifth chapter looks at the system's results, breaking down the findings and showing how they stand up against traditional methods. Finally, the sixth chapter wraps things up. It highlights what the research contributed, discusses the limitations, and suggests what the next steps might look like.

2 – Literature Review

2.1 Foundations of Pay Equity and Algorithmic Analysis

2.1.1 Gender Pay Gap and Persistent Disparities

The pay gap between the two genders also represents a challenge that affects the global community due to occupational segregation and institutional biases occurring in structural discrimination affecting workers (Strittmatter, 2021). Large-scale empirical research also demonstrates that it is true that the current gap in salaries discriminates workers based on gender even after adjusting for occupational differences (Ontario Confederation of University Faculty Associations, 2016). Big data research also suggests the complexity associated with inequality in paying workers despite these biases; it relates to frictions occurring in the labor market affecting women (Bharadwaj et al., 2024).

2.1.2 Limitations of Traditional Pay Analysis Methods

The conventional compensation study typically employs linear regression models where there is assumed to be no change in the underlying relationships between variables—that can be limiting as it may hide complex interaction phenomena occurring in practical workplace scenarios (Fogel, n.d.). The conventional econometric models would also not be able to reveal underlying discriminatory structures because they require predefined structures and have restricted interactivity (Edelsztein et al., 2023). The conventional models essentially ignore compensation disparity dynamics instead of focusing on dynamic skill requirements in present-day organizations (Homolka, 2021).

2.1.3 Emergence of AI in Compensation Systems

Machine learning and AI technology in compensation tools have ushered in a new era of compensation analysis and planning for companies globally. AI models automatically make compensation recommendations, and monitor equity of compensation Recommendation equities for comp across business + Ops (Erica Kochi, 2020). The extended use of compensation algorithms in companies leads explaining the increasing potential for higher accuracy and flexibility in payment decisions. Yet, companies are concerned about bias and lack of transparency in the decisions rendered by AI-technologies-based compensation tools. Recent advancements on transparent AI work to address the problem of bias and transparency in compensation algorithm tools and models (Conitzer, 2023).

2.1.4 Key Fairness Concepts in Pay Equity Analytics

Equitable Pay equity analytics is grounded in theory from computer science, ethics and legal perspectives. Formal and substantial justice Scholars have suggested that formal fairness (which focuses on ensuring like cases lead to like outcomes) should be distinguished from moral or substantive justice, where issues of social injustice based in prejudice are built into the data and institutions (Green, 2020). The theoretical literature on algorithmic fairness underscores the basic trade-offs between equal predictive performance and equal predictive outcomes over all groups (Hellman, 2020). Concepts such as demographic parity (dp), equality of odds (eods) and counterfactual fairness are proposed to guarantee fairness awareness in remuneration modeling, which however cannot be solved by any individual concept (Barocas et al., 2020).

2.2 AI Approaches for Detecting Compensation Disparities

2.2.1 Machine Learning Models for Wage Prediction and Analysis

The application of machine learning models in predicting wages has seen widespread adoption due to their capacity to address complex interrelations beyond human capabilities (Forshaw & Schaffer, 2023). Random forest models and neural networks have been applied in creating more precise models to determine wages (Forshaw & Schaffer, 2023). The application of ML models allows for highlighting complex patterns associated with wages related to job-family memberships and organizations (Brieland, 2020). Injustice related to equity compensation can be addressed by highlighting workers who earn less than their projected amounts (Anderson, 2023). The application of ML models provides increased precision but not necessarily equity unless equity constraints are applied.

2.2.2 Bayesian and Probabilistic Models in Pay Equity

The application of Bayes models yields a versatile probabilistic paradigm for assessing pay equity in terms of assessing uncertainties for pay estimates while leveraging prior knowledge. The utilization of Bayesian hierarchical modeling permits researchers to investigate individual-level variables in conjunction with team-level variables and job-level variables to mitigate biases for comparisons related to pay equity (Cesar, 2020). The application of Bayesian reasoning also enabled identifying discrimination via posterior distributions describing differences for demographically distinct groups (Luo, 2025). The increasing attention towards the application of Bayesian fairness modeling emphasizes its usefulness in compensation audits associated with equity.

2.2.3 Algorithmic Fairness Techniques in Compensation Systems

Fairness-aware models attempt to alleviate bias in forecasting models to recommend just compensation amounts via pre-processing, in-processing, and post-processing techniques. Some techniques applied in pre-processing models aimed at limiting past biases in data; such models form part of fairness strategies (Skanderson, 2021). In-processing techniques combine fairness constraints with learning models to ensure fair compensation (Dolata, 2021). The post-processing techniques mainly ensure fairness in models to provide equality among demographics (Greenlining Institute, 2021). The strategies provide organized platforms for organizations to adhere to fair compensation algorithm models; all strategies have their pros and cons in regards to fairness, interpretability, and accuracy.

2.3 Algorithmic Bias, Data Quality, and Equity Risks

2.3.1 Bias Embedded in Compensation Data

The data used to determine worker compensation can be saturated in ways perpetuating past biases from processes such as promotion decisions, performance management, and jobs allocated to workers (Nachbar, 2022). The increased likelihood of bias from embedded systems such as performance decisions based on group averages to estimate worker productivity adds to systems perpetuating discriminatory compensation decisions (Tilcsik, 2021). Imbalances in worker representation for specific jobs can therefore cause systemic generalizations from algorithmic systems to discriminate workers from underrepresented groups (Madingwaneng, 2024). It must be acknowledged here that workers and compensation data have sociotechnical roots.

2.3.2 Dataset Auditability and Quality Challenges

High-quality data is critical to fairness-sensitive compensation models; nevertheless, organizations face challenges related to data incompleteness, noise, and inconsistencies. The audit framework stresses the importance of checking for demographic data incompleteness and consistency as well as labels' validity to proceed with developing models (Iñigo de Miguel Beriain, 2022). Inadequate data from HR systems can potentially disrupt fairness analyses, particularly if data related to sensitive attributes is incomplete and not accurately recorded (OECD, 2024). Insufficient data in the HR systems can undermine fairness analyses especially when the information of sensitive attributes is incomplete and not properly documented (OECD, 2024). Faulty audit systems may lead AI COMP models to return incorrect findings, and cover rather than reveal discriminatory issues.

2.3.3 Discrimination Risks Posed by Algorithmic Wage-Setting

Algorithmic discrimination happens when discriminatory features, training data samples distribution or goals lead machine learning systems to reproduce and amplify the discrimination.

Automatic compensation systems could perpetuate gender and racial discrimination in wages, if these measures were to be optimised for on the basis of pseudo-sensitive proxies (World Economic Forum, 2020). Additionally, the opacity of black box models further complicates accountability for discriminatory behaviors of systems (Conitzer, 2023). More empirical evidence also indicates that profit maximization algorithmic systems may not take into account fairness and thus increasingly threaten business enterprises (Skanderson, 2021).

2.4 AI Techniques for Detecting Compensation Disparities

Modern compensation systems have become increasingly complex, and traditional statistical tools alone are no longer sufficient for uncovering the structural and hidden drivers of wage inequity. As shown throughout the literature, linear models often fail to capture nonlinear interactions, hierarchical patterns, and historical biases embedded in organizational data (Edelsztein & Waisbrot, 2023; Cesar, 2020). Because pay disparities can emerge from subtle role assignment patterns, departmental differences, and long-term organizational dynamics, more advanced analytical approaches are required to detect them accurately. AI-driven pay equity analytics directly address this challenge by introducing machine learning, Bayesian reasoning, and fairness-aware modeling techniques capable of uncovering inequities that classical methods overlook. The following subsections examine these technologies and demonstrate how they strengthen an organization's ability to identify compensation disparities early, reliably, and in greater depth than traditional tools.

2.4.1 Machine Learning Wage Prediction Models

Machine learning wage-prediction models allow organizations to identify compensation inequities by estimating what an employee should be earning based on objective, job-related characteristics. Unlike traditional regression, which assumes linear relationships between variables, machine learning methods can detect nonlinear interactions and subtle pay patterns that remain hidden under traditional analysis (Edelsztein & Waisbrot, 2023). The central idea is to build a predictive model of “fair salary” and compare this estimate to the actual compensation the employee receives. If the employee's actual pay falls significantly below the model's predicted value, the gap represents a potential equity concern.

The effectiveness of machine learning in uncovering pay disparities has been demonstrated by Edelsztein and Waisbrot (2023), who used an advanced ML decomposition method to analyze wages in a large IT workforce. Their findings revealed that even after controlling for legitimate characteristics such as job level and experience, a statistically significant portion of the gender wage gap remained unexplained. This “unexplained” component signals structural inequities and shows how ML models can identify where interventions may be required.

Further, ML models can detect risks earlier than traditional audits because they can be updated continuously as new compensation data becomes available. This real-time capability prevents small inequities from evolving into systemic disparities and makes ML particularly suitable for dynamic compensation environments (*OECD, 2024*).

2.4.2 Bayesian and Probabilistic Fairness Models

While machine learning predicts expected salaries, Bayesian models extend this analysis by quantifying uncertainty and modeling compensation across multiple organizational layers. Bayesian hierarchical models incorporate employee attributes, job families, and departmental structures to determine whether specific units contribute disproportionately to inequity (*Cesar, 2020*).

Cesar (2020) demonstrated how Bayesian reasoning can uncover subtle discrimination that might not be detected through classical regression. By generating posterior distributions of wage differences, Bayesian models show not only the existence of inequity but also the strength of evidence supporting it. This helps HR decision-makers prioritize which disparities require immediate action. Bayesian methods are also well suited for smaller datasets, where classical models fail due to instability or missing demographic variables (*Cesar, 2020*).

Additionally, Bayesian models handle noisy or incomplete HR data more effectively than conventional approaches. Their explicit modeling of uncertainty offers more reliable estimates of discrimination and is therefore valuable for organizations needing defensible and transparent equity analyses (*OECD, 2024*).

2.4.3 AI Fairness and Counterfactual Techniques

Fairness-aware AI techniques represent some of the most powerful tools for detecting inequities in compensation systems. Counterfactual fairness models evaluate whether an employee's salary would change if their demographic attributes—such as gender—were hypothetically altered while keeping professional qualifications constant (*Ghassemi & Nandigam, 2024*).

These techniques also reveal hidden proxies—variables that appear neutral but reflect gendered or demographic patterns, such as historical role assignments or managerial clustering (*Ghassemi & Nandigam, 2024*). By simulating what an employee “would earn” under counterfactual conditions, organizations can directly test for discrimination even when sensitive attributes are not explicitly included in pay models.

Fairness-aware AI is especially effective for detecting structural and historical inequities. These models assess the entire compensation ecosystem rather than isolated outcomes, exposing patterns linked to organizational history or systemic workforce segmentation (*OECD, 2024*).

2.5 AI Methods for Correcting Compensation Inequities

Detecting compensation disparities is only the first step toward building an equitable salary system; organizations must also apply methods that actively correct the inequities revealed by AI analysis. Traditional HR interventions often lack consistency, scalability, and defensibility, especially when addressing complex or historically rooted wage gaps. AI-driven correction techniques fill this gap by embedding fairness constraints, automated adjustments, and debiasing strategies directly into compensation decision processes (*Ghassemi & Nandigam, 2024; Lakhamraju, 2025*). These methods transform diagnostic insights into concrete, fairness-aligned actions—ensuring that compensation models do not simply identify disparities but also help eliminate them. The subsections below outline how fairness-constrained optimization, pre-processing and in-processing debiasing, and post-processing corrections work together to create compensation systems that are both accurate and substantively equitable.

2.5.1 Fairness-Constrained Compensation Optimization

AI also plays a corrective role through fairness-constrained optimization, which integrates fairness metrics directly into model objectives. Instead of maximizing predictive accuracy alone, these models penalize demographic disparities during training (*Ghassemi & Nandigam, 2024*).

This approach prevents AI compensation systems from reinforcing historical biases. For example, if women or minority groups were historically concentrated in undervalued roles, fairness-constrained optimization ensures those patterns do not influence future salary recommendations. As a result, equity becomes a built-in feature of model behavior rather than an external correction.

Such systems allow organizations to eliminate pay gaps while maintaining operational efficiency, and they provide transparent, repeatable fairness-aligned salary decisions (*OECD, 2025*).

2.5.2 Pre-Processing Bias Reduction Techniques

Since many inequities originate from biased historical data, AI systems use pre-processing techniques to correct issues before training. These include reweighting samples, removing proxy

features, and balancing demographic distributions. *Edelsztein & Waisbrot (2023)* emphasize that biased or incomplete data produce distorted compensation predictions, even when models are technically advanced.

Pre-processing ensures the model learns from data that more accurately reflects equitable practices, reducing the risk of perpetuating structural discrimination. This is especially important when historical HR records encode inequalities in promotion or role assignment (*OECD, 2024*).

2.5.3 In-Processing and Post-Processing Fairness Corrections

Beyond data repair, fairness measures are applied during and after model training. In-processing methods—such as adversarial debiasing—actively prevent discriminatory patterns from being learned (*Ghassemi & Nandigam, 2024*).

Post-processing adjustments modify salary predictions after training. *Lakhamraju (2025)* explains that post-processing can raise predicted salaries for historically underpaid demographic groups or enforce fairness thresholds across the entire organization. Post-processing is particularly useful during annual compensation cycles where fast, defensible interventions are needed.

2.6 Practical AI Applications for Achieving Pay Equity

2.6.1 Real-Time Pay Equity Monitoring and Dashboards

AI-enabled dashboards provide organizations with continuous oversight of salary structures, enabling early identification of developing inequities. Unlike traditional compensation audits—typically conducted annually or semi-annually—real-time monitoring updates as new employee, job, and market data enter the system. This gives HR leaders a dynamic, always-current view of pay distributions across gender, department, seniority level, and job family (*Lakhamraju, 2025*).

These monitoring systems integrate machine learning models that automatically scan for anomalies such as sudden pay compression, wage drift, or deviations from predicted salary benchmarks. By applying embedded fairness metrics, dashboards can warn organizations when disparities exceed predetermined thresholds, allowing early intervention instead of retrospective correction (*OECD, 2024*). Continuous surveillance also reduces the risk that biased managerial decisions accumulate into structural inequities over time.

Dashboard-based monitoring significantly improves organizational accountability. Each flagged case surfaces evidence for decision-making: the predicted salary, the fairness-adjusted benchmark, the employee's actual pay, and relevant historical patterns. This transparency supports more consistent decision-making across units, mitigating the risk that one division unintentionally perpetuates pay disparities more than others (*Lakhamraju, 2025*). It also increases trust among employees, since fairness interventions are triggered through transparent, auditable processes rather than subjective managerial judgment.

2.6.2 Automated Compensation Correction Systems

Automated salary correction systems extend monitoring by recommending or applying adjustments where inequities are detected. These systems aggregate model predictions, fairness constraints, and compensation histories to present HR with evidence-based recommendations. For example, when an employee's pay is systematically below the ML-predicted benchmark after controlling for job-related factors, the system highlights the discrepancy and proposes correction ranges (*Anderson, 2023*).

Anderson (2023) demonstrates the real impact of such mechanisms: in one organizational case study, analytics-driven interventions systematically identified underpaid employees across demographic groups, reducing the gender wage gap over time. Rather than leaving salary adjustments to subjective human assessments, the automated system produced uniform, repeatable corrective actions aligned with organizational fairness standards.

automated correction systems frequently incorporate fairness-aware optimization methods—ensuring that new salary adjustments do not inadvertently introduce new inequities. These systems use constraints developed through fairness literature, such as counterfactual tests or demographic parity thresholds, to guarantee that compensation recommendations remain aligned with equity goals (*Ghassemi & Nandigam, 2024*).

Because automated corrections generate consistent, traceable interventions, they support defensibility in processes subject to regulatory or legal examination. They also allow companies to intervene at scale, something difficult to achieve when relying solely on human review (*OECD, 2025*).

2.6.3 Scenario Simulation and Policy Testing

Beyond identifying and correcting inequities, AI systems allow organizations to simulate compensation policy changes before implementing them. This is especially important because new compensation plans—such as revised salary bands, role restructuring, hiring policy adjustments, or new promotion pathways—often produce unintended consequences.

Scenario simulation models test how proposed changes affect wage distributions under different demographic structures, forecasting whether disparities will widen, remain stable, or shrink (*Lakhamraju, 2025*). These simulations rely heavily on ML prediction and fairness-aware modeling to examine ripple effects of policy adjustments. For example, adjusting salary bands for a technical job family may inadvertently widen gender gaps if women are disproportionately clustered at lower levels of the hierarchy—something scenario models detect before implementation.

Empirical evidence supports the value of predictive modeling for equity-oriented policy design. *Zhang and Wang (2024)* found that AI-driven decision-support systems in Chinese listed firms improved wage fairness for lower-paid employees and reduced overall inequality by identifying inequitable wage structures and enabling organizations to redesign compensation configurations before implementing new policies. Their findings confirm that predictive simulations are not merely theoretical tools but have measurable real-world impacts on improving fairness outcomes and reducing wage disparities.

Scenario-based modeling is also beneficial for anticipating long-term fairness impacts. Because salary structures interact with hiring, promotion, and retention patterns, changes made today may introduce inequities years later. AI simulations allow companies to model these time-delayed effects, reducing the risk of perpetuating structural inequities (*OECD, 2024*).

2.6.4 Integration with Governance, Legal Requirements, and Ethical Standards

AI's practical role in achieving pay equity extends beyond detection and correction; it also helps organizations operationalize regulatory and ethical requirements. The OECD's guidelines emphasize transparency, accountability, data quality, and fairness as mandatory components of AI-enabled HR systems. These requirements can be encoded directly into compensation algorithms, ensuring the model automatically conforms to governance principles (*OECD, 2024; OECD, 2025*).

Governance-integrated AI systems enforce restrictions on proxy variables, document decision pathways, and provide audit logs showing how salary recommendations were made. This reduces

the risk of opaque “black-box” compensation decisions, which could expose organizations to regulatory challenges or ethical concerns. Integrating governance within technical systems also ensures fairness is enforced consistently—rather than depending on individual managers or periodic reviews.

Furthermore, AI governance frameworks help organizations address structural inequities—issues that go beyond numerical fairness metrics. Fairness-aware modeling techniques, such as those described by *Ghassemi & Nandigam (2024)*, promote substantive rather than merely formal fairness by evaluating how demographic or historical patterns influence compensation. These methods align with broader legal and ethical commitments to reduce systemic discrimination.

Finally, governance-aligned AI supports transparent communication with employees. By providing explainable outputs and clear justifications for salary decisions, organizations can increase trust in compensation processes and improve perceptions of procedural justice (*Lakhamraju, 2025*). In this way, AI becomes not only a technical tool but a mechanism for sustaining ethical compensation governance across organizational and regulatory contexts.

3. Conclusion

The research reviewed in this paper demonstrates that AI not only detects compensation inequities with far greater precision than traditional methods, but also provides concrete, systematic mechanisms for correcting them. Machine learning wage-prediction models reveal hidden, nonlinear pay disparities by generating counterfactual “fair salary” estimates and identifying employees whose actual compensation falls below objective, job-related expectations. Bayesian hierarchical models further strengthen this process by quantifying uncertainty, modeling multilevel organizational structures, and isolating whether disparities originate from individuals, specific business units, or structural conditions. These capabilities directly address long-standing analytical limitations that previously obscured the true drivers of wage inequity.

Beyond detection, fairness-aware AI methods actively **solve** pay disparities by embedding equity constraints into compensation systems. Counterfactual fairness, adversarial debiasing, and fairness-constrained optimization prevent algorithms from reproducing historical discrimination and ensure that model outputs remain consistent across demographic groups. Pre-processing, in-processing, and post-processing debiasing techniques integrate fairness at every stage of the analytics lifecycle, transforming AI from a diagnostic tool into a corrective one. These methods provide reliable and repeatable pathways for removing discriminatory influence from organizational data, model structure, and predictive outputs.

AI also enables long-term, institution-level correction through practical applications such as real-time equity dashboards, automated salary-adjustment workflows, and scenario simulation engines. These systems monitor disparities continuously, propose or enact targeted salary corrections, and allow organizations to test new compensation policies before implementation—ensuring that equity is maintained not just at a single point in time but across future workforce cycles. In doing so, AI operationalizes fairness in daily HR processes and strengthens organizational accountability, transparency, and regulatory compliance.

Together, these capabilities form a comprehensive framework through which AI both *identifies* and *remediates* wage inequities. By combining advanced prediction, probabilistic reasoning, fairness engineering, automated correction, and governance-aligned design, AI enables compensation systems that are not only more accurate but substantively more equitable. As a result, AI becomes a direct mechanism for solving pay disparities rather than reinforcing them—supporting fair, transparent, and accountable compensation practices suited for the modern workplace.

3. Research Methodology

3.1 Research Design: Design Science Approach

This study uses the Design Science Research (DSR) methodology as its main approach because DSR fits the goal: building and evaluating new tech solutions for real-world challenges. DSR pushes researchers to create purposeful artifacts targeting specific organizational problems, while also feeding academic knowledge through systematic design, implementation, and evaluation. For this study, that artifact is an AI-based system for compensation fairness—built to spot, assess, and fix pay inequities in organizational data.

Where theoretical or straight empirical methods mostly analyze or describe, DSR is all about solving problems by building actual, usable systems. That’s why it’s a natural fit for AI-driven pay equity analytics. We don’t just want to uncover pay gaps—we want tools that help close them. With DSR, you build, test, and refine artifacts in cycles, which matches the research goal: turn high-level fairness ideas into something HR can actually use in decision-making.

DSR also lets you blend complex components—machine learning models, fairness metrics, simulations, interfaces—into one working system. Tackling compensation inequity needs that kind of complete, “big picture” approach. Just looking at one analytic angle won’t cut it. By using DSR, this research stays grounded in real-world needs while keeping academic rigor.

3.2 Design Science Process Steps

This study follows six DSR stages: problem identification, objective definition, design and development, demonstration, evaluation, and communication. These steps keep the artifact development orderly and let us check that the system works in the real world.

3.2.1 Problem Identification

First, we define the problem. Organizations struggle to pay people fairly, thanks to both old-school analytics—unable to catch nonlinear patterns, hidden biases, or complex disparities—and workforce structures that keep getting more complicated. Even with AI in the mix, new problems arise, like algorithmic bias or a lack of transparency. All of this underlines the need for a system that not only detects pay gaps but also supports fairness, explainability, and accountability.

3.2.2 Objective Definition

With the challenge in focus, next comes clear objectives. The main goal: build an AI system that predicts fair salaries, detects inequities, finds bias, and actually helps fix pay gaps. The system delivers more than just data; it gives organizations specific recommendations for action. Other objectives include clear, explainable outputs and simulation tools, so users can try different scenarios. These guide how we design and develop the system, making sure it stays useful for both practice and research.

3.2.3 Design and Development

Here the artifact comes to life—an interactive, AI-powered compensation fairness app. The core is a regression-based machine learning model predicting fair pay based on traits like experience, performance, or job role. We then compare predicted salaries to actual ones and flag the differences.

But the system does more. There are modules for detecting pay inequities, analyzing bias, visualizing data, running simulations, and generating recommendations. Inequity detection highlights employees clearly underpaid vs. the model's predictions. Bias analysis looks for systemic disparities among demographic groups. The simulation engine lets users allocate pay adjustment budgets, focusing resources on those most underpaid. A user-friendly interface ties everything together, making the system accessible for hands-on analysis and decision support.

3.2.4 Demonstration

In this phase, we put the system to work on real employee data—salaries, gender, roles, experience, and performance. The model predicts fair salaries, calculates fairness metrics, and highlights inequities.

The outputs include detailed explanations at the employee level, fairness scores, gender pay gap numbers, and bias detection signals. Users can try out wage adjustments with a set budget and instantly see the impact on overall fairness. This demonstration proves the system's practical strengths and its ability to drive data-based decisions about compensation.

3.2.5 Evaluation

Next, we see how well the system meets its aims. The evaluation focuses on predicted salaries, inequity flags, pay gap measures, and simulation results. We check whether the system pinpoints underpaid employees, uncovers bias, and suggests real actions to improve compensation fairness.

We also run test cases to validate specific functions—salary prediction accuracy, correct flagging thresholds for inequities, reliable bias detection, and the logic behind budget-based pay adjustments. This phase confirms the integrated system works as intended, blending detection with practical remedies for pay inequity.

3.2.6 Communication

The final step: sharing results with both academic and professional audiences. This happens through the documentation of system design, how we built it, how we tested it, and what we learned—summed up in this thesis. These findings highlight how AI-driven tools can close compensation gaps and make fairness analytics a part of day-to-day business decisions.

This work contributes to both theory and real-world practice—offering a framework for building AI compensation systems and a functioning prototype ready for adaptation. By communicating this research, we invite others to push the field further and spread AI solutions for pay equity.

3.3 Justification for Using Design Science

DSR is the right method for this study for one simple reason: compensation inequity isn't just an analytical issue—it's a practice problem. Most research methods focus on analysis and explanation. They rarely deliver a hands-on tool to fix things in real organizations.

Design Science Research is different. It expects you to build something useful—a system that directly addresses the challenge. DSR also lets us fold together machine learning, fairness analysis, and decision support into one artifact, so we're not piecing solutions together in isolation. It supports cycles of improvement, too. As a result, the system is effective and can adapt over time.

DSR fits especially well with AI-driven pay equity analytics, since that field connects computer science, HR, and ethics. The approach bridges theory and practice, delivering a solution that's both rigorous and relevant to organizations aiming for fairer pay. I pin down the central problem. Organizations still struggle with fair compensation, often because traditional analytics

just don't cut it—they miss nonlinear relationships, hidden bias, and structural disparities in salary data.

4. System Design and Implementation

4.1 System Overview

AI-Driven Compensation Fairness Prototype

University demo: explainability, bias detection, and targeted equity correction simulation

Upload Data

Upload CSV or Excel

Upload 200MB per file - CSV, XLSX, XLS

No file uploaded. Using synthetic sample dataset.

Employee_ID	Gender	Role	Salary	Experience	Performance
7	Male	Engineer	63070	3	4
8	Male	Designer	77643	6	3
9	Female	Manager	73042	4	4
10	Female	Designer	61852	5	1
11	Female	Manager	75006	11	1
12	Male	Manager	74020	10	3
13	Male	Engineer	62778	10	1
14	Male	Manager	64186	6	2
15	Female	Designer	158926	15	4
16	Female	Analyst	38929	2	1

This research introduces an AI-powered application built to tackle compensation fairness head-on. At its core, it's a decision-support tool for HR professionals—a platform that blends machine learning, fairness analytics, and simulation tools so users can dive deep into organizational pay structures and expose hidden inequities. HR teams can upload their actual employee data or generate sample sets, then analyze pay distributions, flag outliers where salaries just don't add up, and run simulations for salary adjustments, all under real-world budget constraints.

The way the application's put together makes it pretty flexible. Each part of the system handles its own task. First, the tool processes the employee datasets and applies a machine learning model that estimates fair salaries, using factors like job experience, individual performance, or the employee's role. These estimates then go head-to-head with real salary figures, making it easy to spot where compensation isn't matching up—essentially, shining a light on potential inequities that might otherwise stay buried. But it doesn't stop at just detecting gaps. The system

adds another layer with advanced bias detection, detailed fairness metrics, and straightforward visualizations. Users don't need to wade through endless spreadsheets—they get clear, interpretable graphics that break down where disparities exist and how big the gaps are.

A standout feature is the built-in simulation engine. Organizations can experiment with different strategies for adjusting salaries—say, they come up with a fixed budget for pay corrections. The system sorts employees to find those most underpaid, then adjusts salaries accordingly. Users see exactly how these changes would affect the broader fairness landscape across the company. Bringing detection, rigorous analysis, and simulation together, this tool gives organizations something concrete: a practical, actionable way to spot compensation gaps and test out solutions. It's not just academic theory—it's designed to make a real impact on fairness and equity in organizational pay structures.

4.1.1 Model Selection and Justification

Dashboard Metrics

Average Salary
\$70,789

Gender Pay Gap
10%

Flagged Inequities
40

Underpaid Employees
30

View employee fairness table

Employee_ID	Gender	Role	Experience	Performance	Salary	Predicted_Fair_Salary	Delta_%	Potential_Inequity
1	Male	Manager	15	1	97528	82402	18	🔴
2	Female	Designer	3	3	57256	57626	-1	🟡
3	Male	Designer	8	1	60231	62742	-4	🟡
4	Male	Manager	11	4	82215	85129	-5	🟡
5	Female	HR	8	4	56803	76868	-26	🔴
6	Male	Manager	7	1	65110	58822	9	🔴
7	Male	Engineer	3	4	63070	62569	1	🟡
8	Male	Designer	6	3	77643	66367	17	🔴
9	Female	Manager	4	4	73942	65388	12	🔴
10	Female	Designer	5	1	61882	54281	14	🔴

We looked at several machine learning models while designing the system—decision trees, random forests, even neural networks. In the end, we went with linear regression. The reasoning wasn't just technical; it came down to transparency and interpretability. HR teams need clear, defensible explanations for compensation decisions, and the last thing anyone wants in that setting is a model that spits out numbers but can't explain itself. Sure, complex models might squeeze out higher accuracy, but they tend to be opaque, making it tough to justify their recommendations to employees or management. Linear regression felt like the right compromise: it offers solid predictive capability without turning into a black box. This choice fits perfectly with the system's aim—helping HR make compensation decisions that are fair, accountable, and fully transparent.

At the heart of the system, linear regression does all the heavy lifting for salary predictions. The model isn't just easy to use—it's plainspoken and fits the kind of structured, tabular data HR

handles every day. Linear regression draws a direct line between salary (the dependent variable) and factors like experience, performance, and job role. It learns these relationships from historical data, so when it generates a salary prediction, it's really giving you an estimate grounded in actual, objective job attributes. In this way, the model reflects expected compensation based on hard facts, not gut feelings.

Transparency is where linear regression really shines. It doesn't just churn out predictions; it shows you, step by step, how every factor influences the outcome. In compensation analysis, this clarity is essential. HR professionals—and employees—need to understand where salary numbers come from, not just accept them blindly. By making these relationships visible, linear regression helps build trust and reinforces accountability. People can see exactly how their experience or performance feeds into their salary, and decision-makers can justify those outcomes with confidence.

Linear regression also gives us a sturdy baseline for fairness analysis. Because the model spells out how input features connect to salary outcomes, we can measure how predicted salaries differ from actual ones. These differences—the deviations—help spot inequities and are key to evaluating fairness across the system. While more advanced models might sharpen the accuracy a bit, the trade-off is usually complexity and loss of interpretability. In HR, that's a risk we don't take. By focusing on models that people can understand and apply in practice, the system delivers what matters most: transparent, fair, and defensible compensation decisions.

4.2 Data Processing and Feature Engineering

View employee fairness table

Employee_ID	Gender	Role	Experience	Performance	Salary	Predicted_Fair_Salary	Delta_
19	Male	Analyst	6	1	62953	57102	
20	Male	Manager	15	3	99080	91767	
21	Male	HR	6	4	63747	72029	
22	Female	Analyst	14	1	75114	76952	
23	Male	Analyst	8	2	56762	67364	
24	Male	Manager	11	5	96569	98772	
25	Male	HR	7	5	67360	78432	
26	Female	HR	4	3	46968	66748	
27	Female	Engineer	12	3	72216	63307	
28	Female	Analyst	15	2	82881	67125	

Potential_inequity

10	
8	
-10	
-4	
-13	
8	
-14	
-23	
-13	
-5	

The system's success is tied to the quality and structure of the data fed into it. In this research, the dataset features employee records packed with attributes directly linked to compensation: salary, gender, job role, years of experience, and performance ratings. We chose these specific features because they're central to how salaries are determined and play a key part in assessing fairness.

Before training the model, we put the data through several cleaning and formatting steps to ensure it's dependable. This means tackling missing values, transforming categorical information like job roles, and adjusting numerical features as needed. One-hot encoding handles the categorical roles—it breaks them down into machine-friendly bits, so the model can recognize the nuances between different job categories.

Feature engineering is more than a technical step here; it sets the tone for both model accuracy and fairness. Factors like experience and performance are primary predictors—they're defensible reasons for salary differences. We treat sensitive variables such as gender with special care, using them strictly to gauge fairness instead of letting them sway predictions. By walling off these attributes, we stop the model from picking up biased compensation patterns, but we still keep the door open for detecting any unfairness through comparative analysis.

4.3 System Functional Modules

This system is structured around a series of functional modules, each tackling a unique piece of the compensation puzzle. Together, they form a robust framework for identifying and addressing pay disparities, adding rigor and clarity to the process.

4.3.1 Salary Prediction Module

The salary prediction module sits at the core, estimating fair salaries using a linear regression model trained on employee attributes. Input factors like experience, performance, and job role feed into the model, producing predicted salary values for each person. These values express what compensation would look like in a fair, objective world.

These predictions don't sit idle—they anchor all further analysis. When we compare predicted salaries to what employees actually earn, the system calculates deviation percentages that immediately highlight who's underpaid or overpaid. This comparison builds a solid, numerical foundation for spotting inequities and guides decision-makers toward fact-based conclusions.

4.3.2 Inequity Detection Module

Next up is the inequity detection module, designed to pinpoint employees whose actual salaries stray too far from their predicted values. The system uses a specific threshold—say, a 5%

difference—to decide when a “deviation” crosses the line. If an employee’s pay falls short of their predicted salary by more than this margin, the system flags them as potentially underpaid.

This method sticks closely to the system’s guiding principle: inequity is a measurable gap between expectation and reality. By automating the flagging process, we ensure that compensation disparities are identified consistently and impartially, removing the guesswork and keeping subjective bias out of the equation. The result is a more transparent, reliable approach to pay analysis and decision support.

4.3.3 Pay Gap Analysis

Fairness Analysis

Select employee(s) for explanation

1

Employee 1 is a Manager with 15 years of experience and performance score 1. The expected fair salary is 82,482; actual salary is 97,528 (+18% vs expected). Potential inequity flagged (15% difference).

Gender Bias Detection

Potential gender bias: females appear more underpaid than males

Gender	Average Delta, %
Female	-2
Male	2

The pay gap analysis module goes beyond basic compensation review by calculating key fairness metrics—most notably, the gender pay gap. It does this by taking the average salaries of different demographic groups and measuring the percentage difference between them. This straightforward comparison reveals where systemic disparities exist within an organization, offering more than just raw numbers; it shines a light on entrenched patterns that otherwise go unnoticed.

With this module, decision-makers can see how compensation trends play out across the entire company. It captures the big picture, showing macro-level patterns that shape the pay structure. This helps put the more granular findings from the inequity detection module into context. That module zooms in on individual cases—maybe a specific employee’s wage compared to a peer—while the pay gap module frames the larger narrative.

By weaving together both levels of analysis, the system doesn’t just highlight isolated problems or sweeping trends—it draws a complete, nuanced map of pay equity within the organization. Decision-makers get the context they need for meaningful action, equipped to tackle inequities

both in broad strokes and at the level of individual employees. It's a more holistic strategy to foster fairness, using data to drive both oversight and change.

4.3.4 Bias Detection Mechanism

Root Cause Analysis

Negative Delta_RL means actual salary is below model-estimated fair salary.

Role-Based Analysis

Role	Employees	Avg_Salary	Avg_Predicted_Fair_Salary	Avg_Delta_Percent
HR	10	65204	65409	-0.3
Analyst	8	65942	70642	-6
Engineer	11	67191	69132	-3
Designer	12	71190	70612	2
Manager	17	80191	74711	6

Must understand role HR (Avg Delta: -0.3%)

Experience-Level Analysis

Experience_Level	Employees	Avg_Salary	Avg_Delta_Percent
0-3 yrs	9	65201	-1
4-7 yrs	21	67190	3
8-12 yrs	22	70606	-2
13+ yrs	8	84639	-1

Must understand group 8-12 yrs (Avg Delta: -2%)

Performance Analysis

Performance	Employees	Avg_Salary	Avg_Delta_Percent
1	10	65322	3
2	8	67193	-7
3	18	70605	2
4	12	77612	5
5	7	73675	-6

The bias detection module looks for patterns that link salary differences to demographic traits. It digs into the numbers, comparing how much compensation deviates for groups like men and women. By tracking these percentages, it shows whether pay gaps reliably line up with who people are.

When it spots a real difference—say, a consistently higher deviation for one gender—it flags the finding as a bias signal. That's a heads-up for possible discrimination. With this approach, organizations not only catch structural inequities, but they get a practical tool for digging deeper. It prompts them to examine why these disparities exist, encouraging them to tackle the root causes rather than just surface symptoms. The process, in effect, turns raw data into actionable steps toward fairer compensation.

4.3.5 Budget Simulation Engine

Simulation (Budget Allocation)

Budget is allocated automatically to the most underpaid employees first.

Total Adjustment Budget (\$)

\$50000

How this works: The system uses a machine learning model to estimate a fair salary for each employee based on experience, performance, and role.

When you enter a budget, the system automatically identifies underpaid employees and increases their salaries toward the predicted fair value. Employees with the largest pay gaps are adjusted first until the budget is fully used.

This helps reduce pay inequities and can improve the gender pay gap by prioritizing employees who are most underpaid.

Total Cost of Adjustments

\$50,000

Average Salary

\$70,789

↑ \$833

Gender Pay Gap

8%

↓ -2%

Flagged Inequities

38

↓ -2

Cost vs Improvement Insight: \$50,000 changes gender pay gap from 10% to 8% and flags from 40 to 38.

Adjusted Employees (Explainability)

Employee_ID	Gender	Role	Original Salary	Adjusted Salary	Increase Amount	Increase %	Adjustment Status
5	Female	HR	56603	70341	13738	24	Adjusted
41	Female	HR	37055	50454	13409	36	Adjusted
57	Female	Designer	64099	86952	22853	36	Adjusted
2	Female	Designer	57256	57256	0	0	Not Adjusted
3	Male	Designer	60231	60231	0	0	Not Adjusted
4	Male	Manager	81215	81215	0	0	Not Adjusted
12	Male	Manager	74020	74020	0	0	Not Adjusted
13	Male	Engineer	62779	62779	0	0	Not Adjusted

The budget simulation engine lets users test different strategies for fixing pay gaps by allocating a set amount of money to adjust salaries. It doesn't just randomly distribute funds—first, it finds the employees who are underpaid and ranks them by how much their salaries lag behind. Starting with those with the biggest gaps, the system bumps up their pay one by one, stopping when the budget runs out.

Running these simulations sheds light on how every resource allocation decision shapes overall fairness. The engine isn't just a tool for crunching numbers; it allows organizations to experiment, run scenarios, and see the concrete impact on compensation equity. This kind of analysis makes it easier to develop smart, strategic plans and follow through with adjustments that actually work.

4.3.6 Recommendation System

The recommendation module doesn't just spit out generic advice—it crafts targeted, practical steps based on what the system uncovers. Sometimes it's a call to adjust certain salaries right away. Other times, it signals the need for a closer look at specific roles or departments where things seem off. When patterns of bias show up, the module highlights those as well and suggests avenues for correction.

These recommendations come from a blend of predictive analytics, detection tools, and simulation outcomes. By grounding advice in hard data, the module raises the standard for actionable organizational insights. This makes the guidance trustworthy and gives leaders confidence in their decisions.

4.4 Application of Design Science

Design Science Research shapes every element of this system, moving from theory into hands-on application through a systematic, iterative approach. The real-world challenge—compensation inequity—gets reframed as a system design problem. The goal? Build an artifact that doesn't just spot disparities but also helps close them.

The result is a robust system, the artifact itself, which brings together machine learning, fairness analytics, and simulation features. Using actual datasets, the demonstration highlights how the artifact operates, surfacing inequities and producing real recommendations. Evaluation happens through the system's outputs: fairness metrics, simulation findings, and actionable insights. By tying these results back to the original research problem, the system proves it can actually help address compensation disparities, not just study them.

4.5 Expert Validation (HR Interview)

To enhance the practical relevance of the system, expert validation is conducted through consultation with an HR professional from [Elbaron-group](#). The feedback obtained highlights the importance of transparency, explainability, and usability in compensation systems. The expert emphasizes that HR decision-makers require clear justifications for salary adjustments and must be able to communicate these decisions effectively to employees.

ME : “From your perspective, what is the biggest challenge in managing pay equity?”

HR Manager (Tarek): “One of the biggest challenges is identifying inequities accurately. We often rely on reports and averages, but they don't always show the full picture, especially at the individual level.”

ME: “How do you find the system's inequity detection feature?”

HR Manager (Tarek): “It is very effective. The system highlights employees whose salaries deviate from expected values, which makes it much easier to identify potential issues quickly instead of manually analyzing large datasets.”

ME: “What about the salary explanation feature?”

HR Manager (Tarek): “That’s actually one of the most useful parts. Employees often ask why they are paid a certain amount, and having a clear explanation based on data would improve transparency and trust.”

ME: “Do you think the simulation feature is realistic?”

HR Manager (Tarek): “Yes, especially because it considers budget constraints. In real situations, we cannot adjust all salaries at once, so prioritizing the most underpaid employees is a practical approach.”

ME: “Can this system fully solve the gender pay gap?”

HR Manager (Tarek): “It can definitely help reduce it by identifying and correcting salary differences, but it cannot solve the problem entirely. Factors like hiring practices and promotions also play a role.”

The responses from the HR manager indicate that the system aligns well with real-world HR needs, particularly in improving transparency, efficiency, and decision-making in compensation management.

5. Discussion

5.1 Interpretation of Results

The results from the developed system really underscore how effective AI can be at both spotting and analyzing pay inequities. By relying on a regression-based model to predict what fair compensation should look like, the system goes beyond surface-level statistics. Instead, it zooms in on the gap between what people actually earn and what they ought to earn, given their qualifications. That means the technology can put an actual number on inequity, highlighting employees who aren't being paid what their background and contributions deserve. Traditional compensation analysis often gets bogged down in sweeping averages or time-consuming manual reviews, neither of which can accurately pinpoint who's underpaid. The fact that flagged employees align with the system's defined criteria just reinforces how systematic and reliable this method is.

Going a step further, the system breaks down gender pay gap metrics and deviation patterns across demographic groups, which really opens the door to exploring the roots of structural inequities. Spotting consistent bias signals—especially when one demographic consistently scores lower on predicted-to-actual salary ratios—shows these issues are rarely individual flukes. Instead, the data suggest broader organizational patterns that need attention. This is precisely why it's so important not to focus solely on either individual cases or overall trends. Neither one tells the full story. The dual-level analysis—individual and aggregate—brings a more thorough understanding of compensation inequity.

There's also the practical side, shown clearly in the simulation results. The system isn't just diagnostic; it suggests targeted salary adjustments, showing how these can shrink disparities even within real-world budget limits. When the process prioritizes the most underpaid employees, it guarantees resources make the most impact. This blend of detection and action, with every step structured and transparent, proves AI isn't limited to flagging problems—it's also a solid guide for making fairer compensation decisions.

5.2 Addressing the Research Problem

This study set out to tackle a long-standing problem: traditional compensation analysis tools often fall short, both in pinpointing pay inequities and suggesting concrete steps for correction. The solution proposed here—a system combining machine learning with fairness analysis and simulation—all within a single architecture, presents a significant leap forward.

Where most conventional methods lean heavily on static data or subjective interpretation, this AI-driven system shifts the process to a dynamic and evidence-based footing. Predicted salary benchmarks bring nuance, accounting for each employee's role and qualifications. No more relying on vague "industry averages" or guesswork. Built-in bias detection adds another layer, ensuring persistent, systemic issues aren't swept under the rug.

Crucially, the system moves beyond analysis and offers actionable recommendations. With a simulation engine and guidance module, organizations actually have a plan for closing the gaps once they're identified—a missing piece in much of the existing literature. So, instead of stopping at "here's the inequity," this research delivers a strategy that encompasses detection, explanation, and correction, all integrated seamlessly. It's a roadmap for real progress towards pay equity.

5.3 Comparison with Traditional Methods

The new system steps away from the limitations of classic approaches to compensation analysis, which tend to rely on basic averages, general regression models, or periodic (and often outdated) audits. Those tools often gloss over the subtler forms of inequity hidden in complex datasets and miss how different variables actually interact to influence pay.

AI-driven compensation analysis works differently. By generating tailored salary predictions for each individual, it brings a granular level of clarity that aggregate numbers just can't match. It enables organizations to quickly zero in on specific employees who are underpaid, not just broad trends or group-level disparities. Continuous, real-time analysis means organizations stay in touch with fairness concerns as they evolve—not just when the annual review comes around.

Another important distinction is the system's built-in support for fairness and decision-making. Traditional methods may identify gaps, but they rarely suggest how to close them or help organizations take the next step. Here, automation, simulation, and recommendations co-exist, guiding HR teams in a consistent and transparent way. This transition—from manual, fragmented efforts to a unified, intelligent workflow—marks a real advantage in efficiency and effectiveness.

5.4 System Strengths

Transparency stands out as one of the system's defining features. Because the model used for salary prediction is interpretable, users can see exactly how it weighs qualifications, positions, and other variables. Seeing the reasoning behind a pay recommendation—not just the result—makes the data more meaningful and earns trust from both decision-makers and employees.

Automation offers another layer of strength here. By handling tasks like salary prediction, inequity detection, and bias analysis automatically, the system cuts down on the kind of human error and unintentional bias that often creep into manual reviews. This not only boosts consistency, but also opens the door to large-scale use, where manual audits would be unmanageable.

Unlike approaches that treat fairness as a final afterthought, this system bakes equity into every stage—using established fairness metrics, automatic bias detection, and ongoing correction mechanisms. Monitoring and adjustment aren't side jobs; they're central features, always on.

The simulation engine is particularly practical. Before any policy shifts happen for real, organizations can test out different strategies within the system, seeing the impact in advance.

This means compensation planning becomes more data-driven and less reliant on gut feeling, leaving organizations better equipped to create fairer workplaces.

5.5 System Limitations

Every system has boundaries, and there are limitations worth noting here. First, the size and diversity of the dataset are essential. Smaller or less varied datasets limit how accurately the model can predict fair compensation, and can undermine the broader applicability of the results.

The choice to use a linear regression model, though excellent for interpretability, is also a constraint. Linear regression captures straightforward relationships, but real-world compensation often includes complex, nonlinear patterns. While advanced machine learning models could improve predictive power, they can also turn the system into a “black box,” reducing transparency and making it harder for users to trust the results.

Another limitation comes from the system’s reliance on set thresholds (like the 5% deviation rule) for highlighting pay inequities. These cutoffs, while practical, can leave some subtle inequities undetected or miss factors that only emerge in more nuanced contexts.

A final caveat—features chosen for the model must truly reflect the factors that drive pay decisions. If important determinants aren't included, predictions and fairness assessments could end up incomplete or, worse, misleading. Recognizing these limitations points the way to future improvements and lays a foundation for further research.

6. Conclusion

6.1 Summary of Findings

This research set out to confront a challenging but crucial issue: identifying and correcting compensation inequities via artificial intelligence. The developed AI-driven compensation fairness system proves that machine learning can pinpoint disparities with a level of precision and nuance unavailable to traditional methods. By integrating everything from salary prediction, inequity detection, and bias analysis, to simulation and recommendations, the system provides a truly comprehensive response to the problem of pay equity.

It's clear that AI-driven techniques bring substantial advantages. They don't just heighten the accuracy of findings; they allow for greater scalability and uncover patterns within the data that previously went unnoticed. Most importantly, the system translates insights into action, guiding decision-making for real improvements in organizational pay practices.

6.2 Theoretical Implications

On the theoretical front, this research pushes the boundaries of digital pay equity analysis. Past studies have usually stopped short at exposing disparities. Here, the integration of detection, simulation, and real-time decision support moves the field toward not just identifying unfairness but also systematically correcting it.

Integrating fairness right into the technical core of AI systems marks another important advance. The work highlights that getting the math right matters, but it's not enough; ethical and social impacts must be front and center. By using Design Science Research, the study also models an effective approach for bridging theory and practice—showing how artifact design can inform methodological progress in real-world settings.

6.3 Practical Implications

The practical upside for organizations is significant. With this system, HR teams gain a powerful, user-friendly tool to analyze data, make transparent decisions, and manage equity proactively. By spotlighting underpaid staff, uncovering bias patterns, and simulating changes, the system helps businesses put fairness into everyday practice, not just policy.

It doesn't stop there—accountability improves, too. Outputs are clear and ready to share, supporting both internal justification and external compliance. By tracking fairness metrics live, organizations can spot problems early, before they root themselves deeply in culture or practice.

6.4 Recommendations

Given the demonstrated benefits, organizations should start adopting AI-driven tools for compensation management and fairness checks. These technologies not only improve accuracy and transparency, but also fundamentally reduce bias and the risk of discrimination.

Data quality matters here. Organizations need to make sure the compensation data they use is reliable and free from embedded biases. It's just as important to incorporate explainable fairness metrics—these build trust and enable ongoing accountability. Simulation features also deserve attention, letting leadership test policy changes before rolling them out. That way, salary strategies stay in line with equity goals.

6.5 Limitations & Future Work

No system or study is perfect, and this one is no exception. The work relied on a limited dataset and a relatively basic model. These choices fit the project's goals, but they inevitably narrow how broadly the findings can be applied. The system also can't account for every factor influencing compensation, so there's a risk of some blind spots in real-world situations.

Future research should embrace advanced methods—ensemble and deep learning models, for example—to sharpen predictive accuracy. Bringing in larger, more diverse datasets and real-time data sources would only make the system stronger and more representative.

Next steps should also target improved fairness metrics and more modern bias detection tools, like causal inference and counterfactual analysis. Finally, real-world implementation and long-term studies would help clarify how much the system can shift organizational equity over time, closing the loop between theory, design, and actual impact.

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Appendix :